Xiao Fan

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Education

- **Tongji University · College of Computer Science and Technology** — Shanghai, China
  B.Eng. in Data Science and Big Data Technology; GPA: 4.86/5; Rank: 1/37 — *2022.09–2026.06*
- **Tsinghua University · Tsinghua Shenzhen International Graduate School** — Shenzhen, China
  D.Eng. in Electronic Information, advised by Prof. Zhi Wang — *2026.09–2031.06*

Research Interests

- **Test-Time Adaptation** — *2025.01–Present*
  Test-time model adaptation under data distribution shifts (domain shift) without access to source data.
- **Self-Evolving Embodied Agent** — *2025.05–Present*
  Embodied agents that learn from environmental interaction to continually self-improve.
- **Agent Memory** — *2026.01–Present*
  System that stores and retrieves information from long-term memory to support reasoning and decision-making.

Experience

- **HeyWhale** — Remote
  Data Analyst Intern (Part-time) — *2024.07–2024.08*
  - **Community Support in Data Science**: Provided timely answers and guidance to learners and developers on data analysis, machine learning, statistics, and visualization questions in online communities.
  - **Tutorial and Educational Content Creation**: Designed and authored clear, structured tutorials and guides, translating complex data science concepts into practical, easy-to-follow instructions.
  - **Knowledge Sharing and Engagement**: Enhanced community learning by sharing example code, best practices, and learning paths to improve technical skills and foster engagement.

Publications (* equal contribution, † corresponding author)

- **Semantic-Space Exploration and Exploitation in RLVR for LLM Reasoning** — *2025*
  Fanding Huang*, Guanbo Huang*, **Xiao Fan**, Yi He, Xiao Liang, Xiao Chen, Qinting Jiang, Faisal Nadeem Khan, Jingyan Jiang†, Zhi Wang†
  Findings of the Association for Computational Linguistics: ACL 2026
  Keywords: LLM Reasoning, Reinforcement Learning, Representation Interpretability
  [PDF] [BibTeX] [Code] [Project Page]
- **MoETTA: Test-Time Adaptation Under Mixed Distribution Shifts with MoE-LayerNorm** — *2026*
  Xiao Fan, Jingyan Jiang†, Zhaoru Chen, Fanding Huang, Xiao Chen, Qinting Jiang, Bowen Zhang, Xing Tang, Zhi Wang
  Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence (AAAI, CCF-A, 17.6% overall acceptance rate)
  Keywords: Test-Time Adaptation, Mixture of Experts
  [PDF] [BibTeX] [Code] [Poster]

Selected Honors and Awards

- **Global Campus AI Algorithm Elite Competition** — National 1st Prize
  Challenge: "BIRADS Classification and Feature Recognition in Ultrasound Breast Images" — *2024.11*
- **15th Lanqiao Cup** — National 3rd Prize
  Python Programming — *2025.08*
- **Tongji University Excellent Student Scholarship** — 1st Prize
  Guo Xie Birong Scholarship, 2 out of 756 students — *2023.12*
- **Excellent Graduate of Tongji University** — *2026.06*

Skills Summary (* Expert, † Proficient, ‡ Basic, § Hands-on Experience)

- **Programming**: Python*, C++†, C†, Bash‡, SQL‡, JavaScript§, TypeScript§
- **Frameworks**: PyTorch†, Scikit†, Streamlit†, Next.js‡, Qt Quick‡, TailwindCSS‡
- **Tools**: Pandas†, LaTeX†, Vim†, Tmux†, Matplotlib†, Git‡, Docker‡
- **Languages**: English† (CET-4: 658, Top 1%; CET-6: 608, ∼Top 5%), Japanese§ (Self-Study on Duolingo)
- **Sports**: Men's Pistol* (10m Air, 25m Rapid Fire, 50m Free; Achieved 1st Place (Team) and 4th Place (Individual) at Shanghai City Championship), Badminton†, Golf‡, Indoor Climbing‡

Volunteer Experience

- **Math Takeaway (Campus Public Welfare Organization)** — Tongji University
  Lecturer for Linear Algebra — *2025.09–2025.12*
  - Delivered a Linear Algebra final exam review lecture.
  - Co-developed two chapter-level knowledge summaries and revision sessions.
  - Produced and published recorded revision lectures on BiliBili.